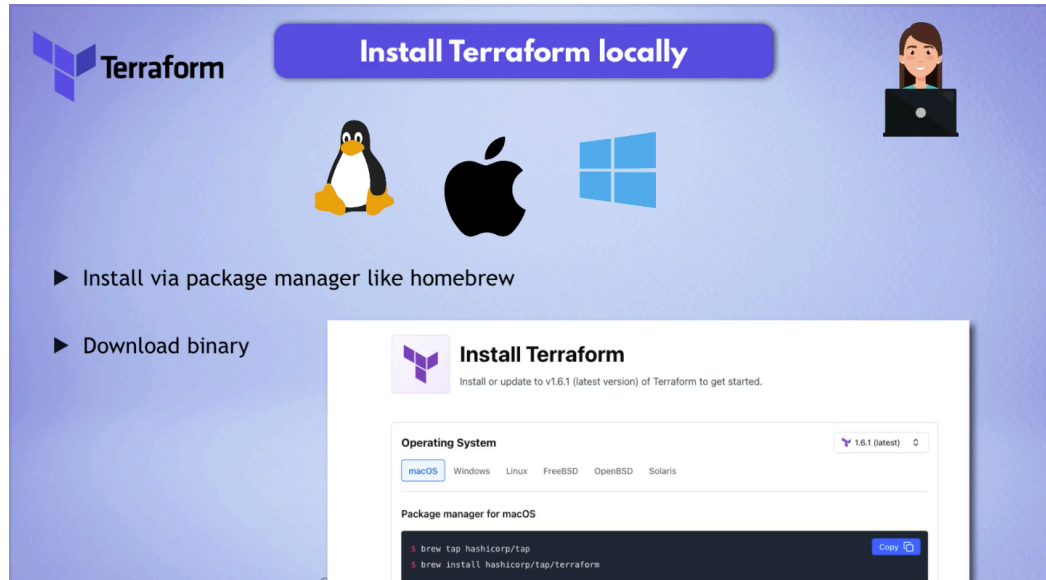


Module 12 - chapter 2

Install Terraform & Setup Terraform Project

The official Terraform installation documentation can be found here:

<https://developer.hashicorp.com/terraform/downloads>



Add TF repo to brew

-> brew tap hashicorp/tap

```
astrid@192.168.1.114 /Users/astrid
% brew tap hashicorp/tap
==> Auto-updating Homebrew...
Adjust how often this is run with `HOMEBREW_AUTO_UPDATE_SECS` or
`HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `HOMEBREW_NO_
✓ JSON API packages.ventura.jws.json
==> Auto-updated Homebrew!
Updated 3 taps (mongodb/brew, homebrew/core and homebrew/cask).
==> New Formulae
ansible@13: Automate deployment, configuration, and upgrading
asccli: App Store Connect CLI to manage apps, versions, and screen
avisynthplus: Improved version of the AviSynth frameserver
baquette: Headless iOS Simulator manager and host-side input inje
```

Execute the brew install command from the official website - to be sure that we get the latest version of TF

-> brew install hashicorp/tap/terraform

```

Tapped 2 casks and 33 formulae (100 files, 1.6MB).
astrid@192.168.1.114 /Users/astrid
% brew install hashicorp/tap/terraform
=> Trusted formula hashicorp/tap/terraform
Warning: You are using macOS 13.
We (and Apple) do not provide support for this old version.
You may have better luck with MacPorts which supports older versions of macOS:
  https://www.macports.org

This is a Tier 3 configuration:
  https://docs.brew.sh/Support-Tiers#tier-3
You can report issues with Tier 3 configurations to Homebrew/* repositories!
Read the above document before opening any issues or PRs.

Warning: The following taps are not trusted:
  mongodb/brew
  weaveworks/tap

Homebrew is currently ignoring formulae, casks and commands from these taps because tap trust is required.
Trust specific formulae, casks or commands with:
  brew trust --formula <user>/<tap>/<formula>
  brew trust --cask <user>/<tap>/<cask>
  brew trust --command <user>/<tap>/<command>
or trust installed formulae from these taps with:
  brew trust --formula mongodb/brew/mongodb-community mongodb/brew/mongodb-database-tools
  brew trust --formula weaveworks/tap/eksctl
You can trust all formulae, casks and commands from these taps with:
  brew trust mongodb/brew weaveworks/tap
Prefer trusting only the specific formulae, casks or commands you need.
Untap them with:
  brew untap mongodb/brew weaveworks/tap
To disable trust checks:
  export HOMEBREW_NO_REQUIRE_TAP_TRUST=1
This is not recommended and will be removed in a later release.
=> Would install 1 formula:
terraform
=> Fetching downloads for: terraform
✓ Formula terraform (1.15.6) Verified 36.5MB/ 36.5MB
=> Installing terraform from hashicorp/tap
  /usr/local/Cellar/terraform/1.15.6: 4 files, 120.9MB, built in 4 seconds
=> Running 'brew cleanup terraform'...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').
astrid@192.168.1.114 /Users/astrid

```

Let's check

-> terraform -v

```

% terraform -v
Terraform v1.15.6
on darwin_amd64
astrid@192.168.1.114 /Users/astrid
%

```

Load Setup

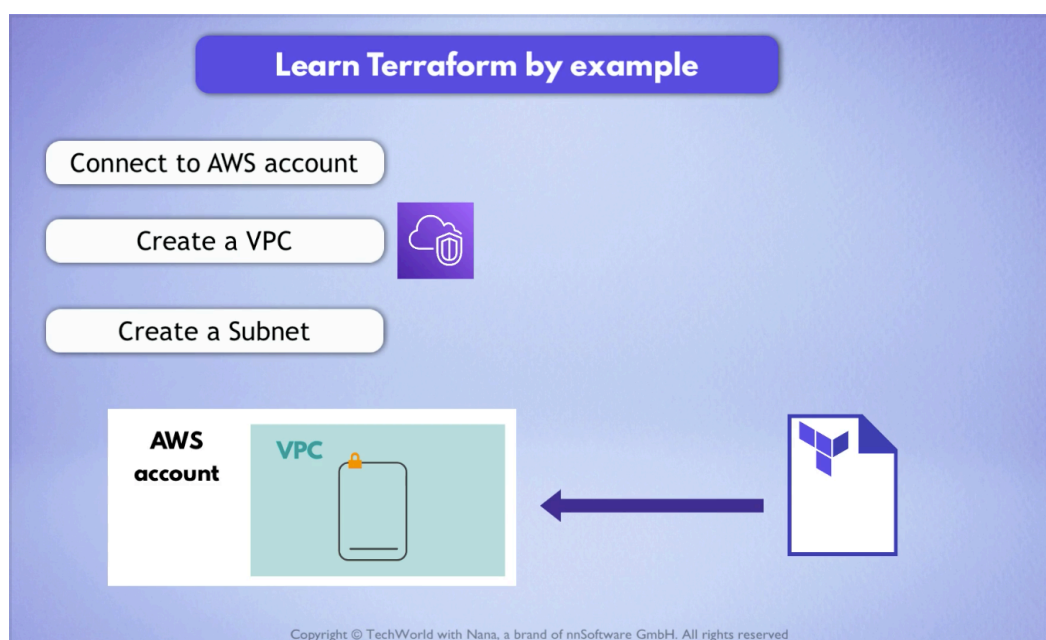
We will use AWS as an example to learn TF concepts. TF is widely use to automate AWS infra.



I will write a TF file that connects to my AWS account and we will create:

- VPC
- Subnet

To learn the basics of TF and its syntax



Create a folder called 'terraform' and we will create files in that folder.

Open terraform folder in IntelliJ and initialise git (this folder is not a repo yet)

-> git init

— tangent, use Oh my ZSH —

My laptop already has ZSH, which is the shell I have been using so far:

```
Last login: Sun Jun 14 11:05:51 on console

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
192:~ astrid$ zsh
astrid@192.168.1.114 /Users/astrid
```

Every time I open iTerm the default shell was BASH so I had to tap 'zsh' to change the shell for that iTerm session.

I changed the default shell to zsh:

-> chsh -s /bin/zsh

```
% chsh -s $(which zsh)
Changing shell for astrid.
Password for astrid:
chsh: /usr/local/bin/zsh: non-standard shell
astrid@192.168.1.114 /Users/astrid/terraform
% bin/zsh
zsh: no such file or directory: bin/zsh
astrid@192.168.1.114 /Users/astrid/terraform
% /bin/zsh
astrid@192.168.1.114 /Users/astrid/terraform
% which zsh
/usr/local/bin/zsh
astrid@192.168.1.114 /Users/astrid/terraform
% chsh -s /bin/zsh
Changing shell for astrid.
Password for astrid:
astrid@192.168.1.114 /Users/astrid/terraform
```

From this git repo <https://github.com/ohmyzsh/ohmyzsh> I get a command to install Oh My Zsh (I have curl so I can use the curl option)

Oh My Zsh is installed by running one of the following commands in your terminal. You can install this via the command-line with either `curl`, `wget` or another similar tool.

Method	Command
curl	<code>sh -c "\$(curl -fsSL https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/install.sh)"</code>
wget	<code>sh -c "\$(wget -O- https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/install.sh)"</code>
fetch	<code>sh -c "\$(fetch -o - https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/install.sh)"</code>

```
-> sh -c "$(curl -fsSL https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/install.sh)"
```

```

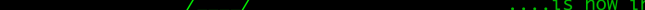
astrid@192.168.1.114 /Users/astrid/terraform
% sh -c "$(curl -fsSL https://raw.githubusercontent.com/ohmyzsh/ohmyzsh/master/tools/install.sh)"
Cloning Oh My Zsh...
remote: Enumerating objects: 1534, done.
remote: Counting objects: 100% (1534/1534), done.
remote: Compressing objects: 100% (1443/1443), done.
remote: Total 1534 (delta 85), reused 1423 (delta 61), pack-reused 0 (from 0)
Receiving objects: 100% (1534/1534), 3.33 MiB | 5.56 MiB/s, done.
Resolving deltas: 100% (85/85), done.
From github.com:ohmyzsh/ohmyzsh
* [new branch]      copilot/fix-broken-theme-jonathan -> origin/copilot/fix-broken-theme-jonathan
* [new branch]      master                               -> origin/master
* [new branch]      rr-13813-introduce-cooldown-feature -> origin/rr-13813-introduce-cooldown-feature
Branch 'master' set up to track remote branch 'master' from 'origin'.
Switched to a new branch 'master'
/Users/astrid/terraform

Looking for an existing zsh config...
Found /Users/astrid/.zshrc.
The existing .zshrc will be backed up to .zshrc.pre-oh-my-zsh if overwritten.
Make sure your .zshrc contains the following minimal configuration if you choose not to overwrite it:
-----
export ZSH="$HOME/.oh-my-zsh"
ZSH_THEME="robbyrussell"
plugins=(git)

source $ZSH/oh-my-zsh.sh
-----
Do you want to overwrite it with the Oh My Zsh template? [Y/n] y
Backing up to /Users/astrid/.zshrc.pre-oh-my-zsh
Using the Oh My Zsh template file and adding it to /Users/astrid/.zshrc.

```

```
Time to change your default shell to zsh:
Do you want to change your default shell to zsh? [Y/n] y
Changing your shell to /bin/zsh...
Password:
Changing shell for astrid.
chsh: no changes made
Shell successfully changed to '/bin/zsh'.
```



Before you scream **Oh My Zsh!** look over the `.zshrc` file to select plugins, themes, and options.

- Follow us on X: [@ohmyzsh](#)
- Join our Discord community: [Discord server](#)
- Get stickers, t-shirts, coffee mugs and more: [CommitGoods Shop](#)

```
→ terraform git:(main) x
```

And as I am CDED into terraform and the folder has been git initiated in IntelliJ

terminal with:

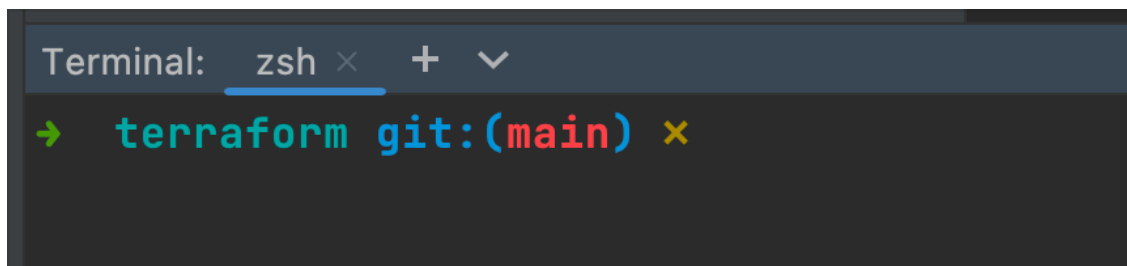
-> git init

We now see

-> terraform git:(main)



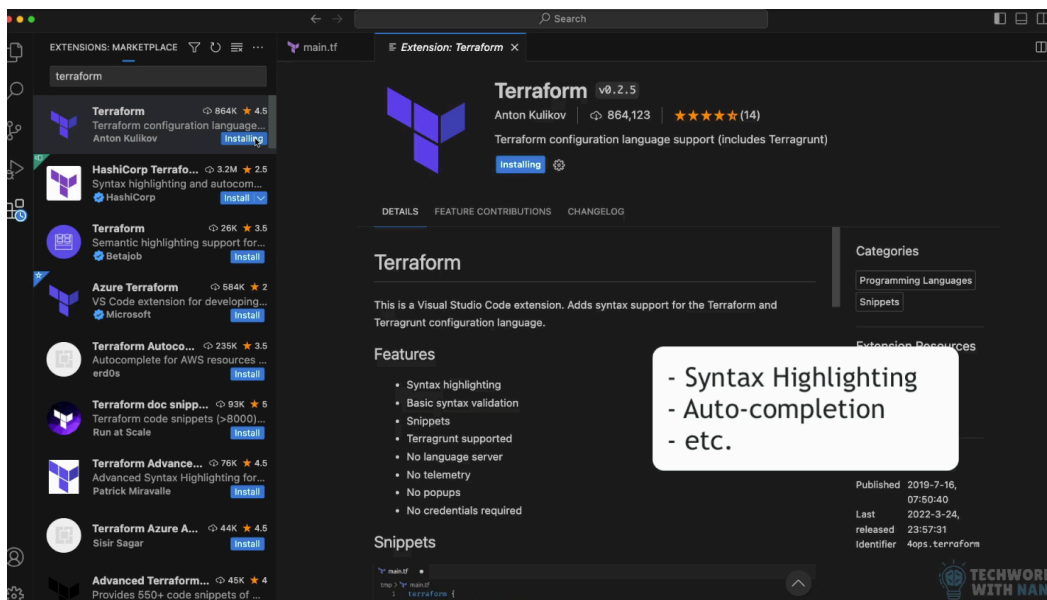
So now I go to IntelliJ and open an zsh terminal window (instead of a bash window) and I have access to Oh my ZSH as well



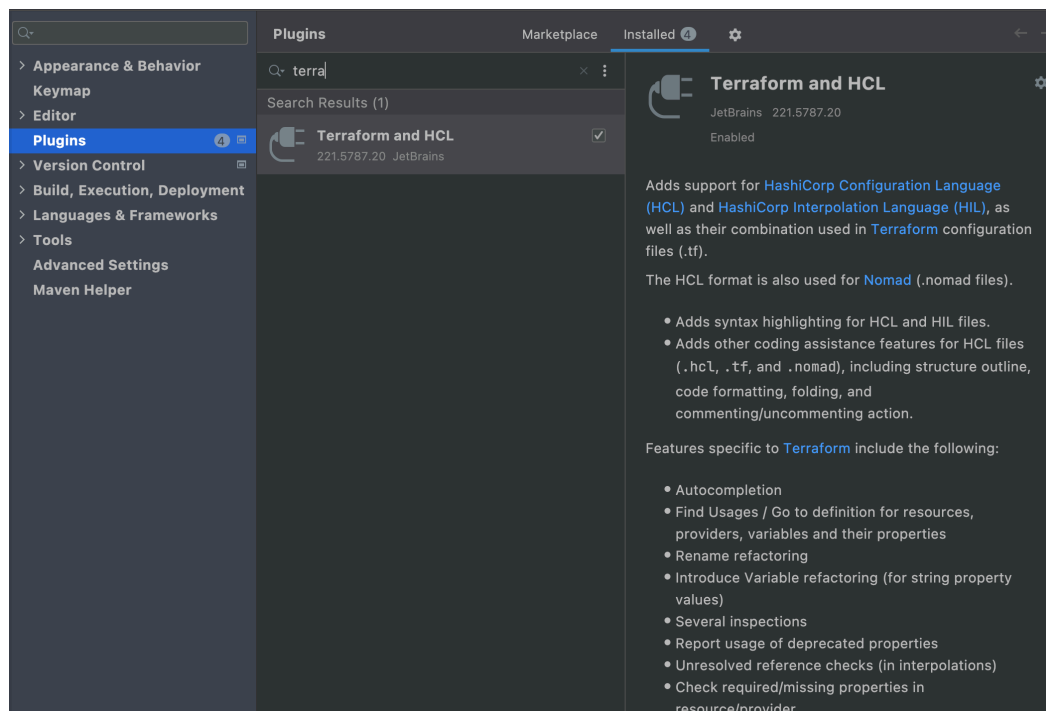
And thanks to Oh my ZSH I can see that -> git init created a main branch and that I am checkout in the main branch ✓

— end tangent —

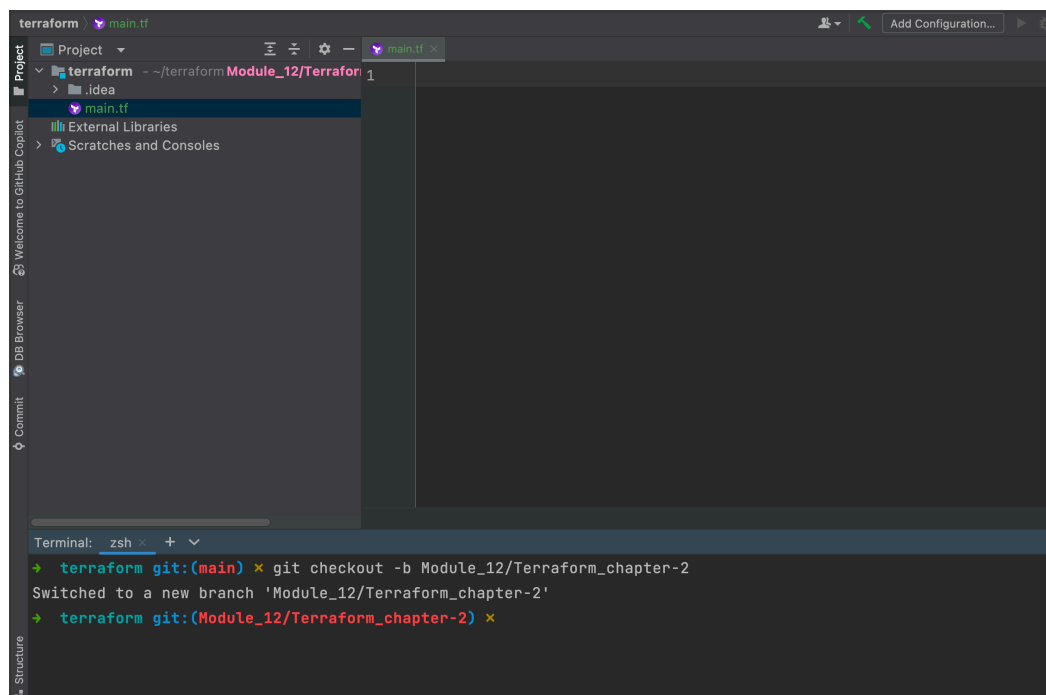
Nana installed a plugin for VSCode which is the IDE she uses



I installed a plugin for IntelliJ instead:



And I created a new terraform file:



— tangent use GitHub CLI —
-> brew install gh

I want to create my GitHub repos using th terminal instead of Github UI


```

Last login: Sun Jun 14 11:37:58 on ttys003
➔ ~ brew install gh
✓ JSON API packages.ventura.jws.json
Downloaded 14.7MB/ 14.7MB
Warning: You are using macOS 13.
We (and Apple) do not provide support for this old version.
You may have better luck with MacPorts which supports older versions of macOS:
https://www.macports.org

This is a Tier 3 configuration:
https://docs.brew.sh/Support-Tiers#tier-3
You can report issues with Tier 3 configurations to Homebrew/* repositories!
Read the above document before opening any issues or PRs.

Warning: The following taps are not trusted:
  mongodb/brew
  weaveworks/tap

Homebrew is currently ignoring formulae, casks and commands from these taps because tap trust is required.
Trust specific formulae, casks or commands with:
  brew trust --formula <user>/<tap>/<formula>
  brew trust --cask <user>/<tap>/<cask>
  brew trust --command <user>/<tap>/<command>
or trust installed formulae from these taps with:
  brew trust --formula mongodb/brew/mongodb-community mongodb/brew/mongodb-database-tools
  brew trust --formula weaveworks/tap/eksctl
You can trust all formulae, casks and commands from these taps with:
  brew trust mongodb/brew weaveworks/tap
Prefer trusting only the specific formulae, casks or commands you need.
Untap them with:
  brew untap mongodb/brew weaveworks/tap
To disable trust checks:
  export HOMEBREW_NO_REQUIRE_TAP_TRUST=1
This is not recommended and will be removed in a later release.
==> Would install 1 formula:
gh
==> Would install 1 dependency for gh:
go
==> Do you want to proceed with the installation? [y/n]

```

After installation, authenticate with:

-> gh auth login

Follow the prompts — it will open a browser to link your GitHub account.

```

➔ ~ gh auth login
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /Users/astrid/.ssh/id_ed25519.pub
? Title for your SSH key: my-laptop-key
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: E8B5-81CB
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
- gh config set -h github.com git_protocol ssh
✓ Configured git protocol
✓ SSH key already existed on your GitHub account: /Users/astrid/.ssh/id_ed25519.pub
✓ Logged in as AstridCaballero
➔ ~ 

```

I already had a ssh key in my laptop that I created to ssh into my DigitalOcean container

```

Last login: Sun Jun 14 12:10:10 on ttys002
➔ ~ cd .ssh
➔ .ssh ls
config          docker-server.pem id_ed25519      id_ed25519.pub  known_hosts     known_hosts.old

```

And I can use the same key for other applications like GitHub.

Also, I execute the below to check if my ssh key is already linked to GitHub:

```
-> ssh ssh -T git@github.com
```

And it is:

```
END OF SSH PRIVATE KEY
> .ssh ssh -T git@github.com
Hi AstridCaballero! You've successfully authenticated, but GitHub does not provide shell access.
> .ssh
```

So when I was asked "Upload your SSH public key to your GitHub account?" I selected the '/Users/astrid/.ssh/id_ed25519.pub' as AI said: it will upload that key to GitHub (it's fine if it's already there, GitHub will just recognize it).

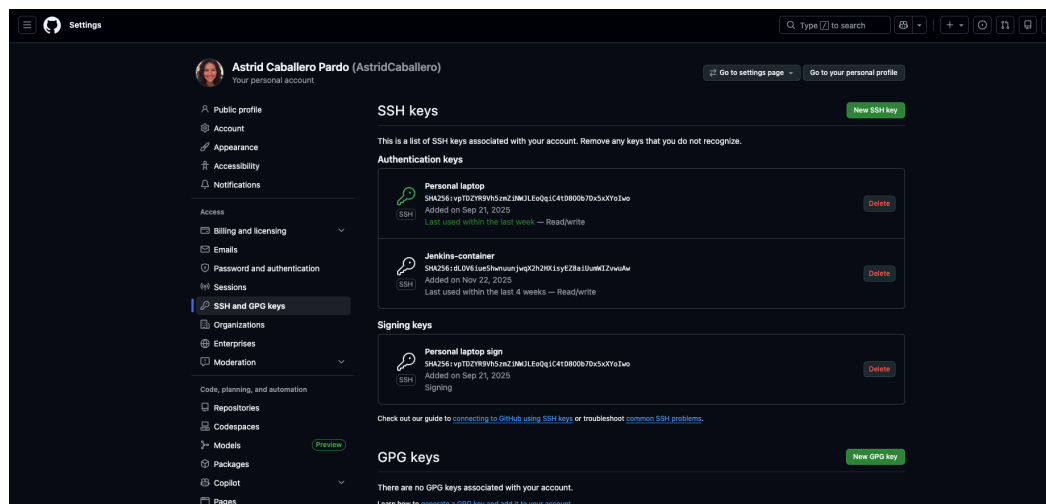
```
? What is your preferred protocol for Git operations on this host? SSH
? Upload your SSH public key to your GitHub account? /Users/astrid/.ssh/id_ed25519.pub
? Title for your SSH key: my-laptop-key
? How would you like to authenticate GitHub CLI? Login with a web browser
```

I check the new key and is not there:

```
-> gh ssh-key list
```

TITLE	ID	KEY	TYPE	ADDED
Personal laptop	132041291	ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA...NLgXDq/tq7dfqssZLJN0AYqQfPVGYNhadSh	authentication	about 8 months ago
Jenkins-container	136694186	ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA...x2xzCu50+gChMt3/53k1n+UbzM9gYz6d5n	authentication	about 6 months ago
Personal laptop sign	679210	ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAA...NLgXDq/tq7dfqssZLJN0AYqQfPVGYNhadSh	signing	about 8 months ago

I check GitHub UI and now I can't see the ssh keys and the new one: 'my-laptop-key' either



AI says: It seems 'gh auth login' recognized the key was already on GitHub and didn't add a duplicate. You're already authenticated and good to go.

IntelliJ is not recognising gh.

-> gh --version

```
→ terraform git:(Module_12/Terraform_chapter-2) × gh --version
zsh: command not found: gh
```

But in my Mac's terminal:

```
→ ~ cd terraform
→ terraform git:(Module_12/Terraform_chapter-2) × gh --version
gh version 2.94.0 (2026-06-10)
https://github.com/cli/cli/releases/tag/v2.94.0
```

If we check IntelliJ's PATH

-> echo \$PATH

```
→ terraform git:(Module_12/Terraform_chapter-2) × echo $PATH
/usr/bin:/bin:/usr/sbin:/sbin
```

The PATH is minimal in IntelliJ, AI says:

IntelliJ's PATH is very minimal, missing **/usr/local/bin** where gh lives.

Your Mac terminal PATH includes **/usr/local/bin** because it loads **~/ .zshrc** on startup. IntelliJ's terminal is not loading it.

I need to edit **./zshrc** file:

-> cat ~/.zshrc

```
→ terraform git:(Module_12/Terraform_chapter-2) × cat ~/.zshrc
# If you come from bash you might have to change your $PATH.
# export PATH=$HOME/bin:$HOME/.local/bin:/usr/local/bin:$PATH

# Path to your Oh My Zsh installation.
export ZSH="$HOME/.oh-my-zsh"
```

This line is commented -> **export PATH=\$HOME/bin:\$HOME/.local/bin:/usr/local/bin:\$PATH**

Uncommenting that line in **~/ .zshrc** will add **/usr/local/bin** to the PATH, so when IntelliJ loads **.zshrc** it will include it. That's the fix we were about to make.

Q:

so it will have **/usr/bin** in the PATH and also **/usr/local/bin**?

A:

After the fix your PATH in IntelliJ will be:

```
/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin
```

And gh will be found.

Q:

/usr/bin is not set in .zshrc, is this done by default?

A:

- Yes, /usr/bin:/bin:/usr/sbin:/sbin are system defaults — macOS sets those automatically for every shell regardless of .zshrc. They're the bare minimum needed to run basic system commands.

.zshrc is where you add extra directories on top of those defaults, like /usr/local/bin for Homebrew-installed tools.

So could help me do the edition and .zshrc file is updated, the line is uncommented:

```
→ terraform git:(Module_12/Terraform_chapter-2) ✕ cat ~/.zshrc
# If you come from bash you might have to change your $PATH.
export PATH=$HOME/bin:$HOME/.local/bin:/usr/local/bin:$PATH
```

And I run

```
-> source ~/.zshrc
```

And now IntelliJ can see gh:

```
-> gh --version
```

```
→ terraform git:(Module_12/Terraform_chapter-2) ✕ source ~/.zshrc
→ terraform git:(Module_12/Terraform_chapter-2) ✕ gh --version
gh version 2.94.0 (2026-06-10)
```

Now I create a new repo and link my local repo to it and I also push:

```
-> gh repo create Module_12_Terraform --private --source=. --push
```

```
+ terraform git:(Module_12/Terraform_chapter-2) gh repo create Module_12_Terraform --private --source=. --push
✓ Created repository AstridCaballero/Module_12_Terraform on github.com
https://github.com/AstridCaballero/Module_12_Terraform
✓ Added remote git@github.com:AstridCaballero/Module_12_Terraform.git
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 4 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (13/13), 5.00 KiB | 2.50 MiB/s, done.
Total 13 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:AstridCaballero/Module_12_Terraform.git
 * [new branch]      HEAD -> Module_12/Terraform_chapter-2
Branch 'Module_12/Terraform_chapter-2' set up to track remote branch 'Module_12/Terraform_chapter-2' from 'origin'.
✓ Pushed commits to git@github.com:AstridCaballero/Module_12_Terraform.git
```

New repo created in GitHub using GitHub CLI -> gh

